

SHEN ZHENG

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4804 Forbes Ave, Pittsburgh, PA 15213

EDUCATION

Carnegie Mellon University, U.S.

Ph.D. in Robotics, Robotics Institute

2024.8 – present

Advisor: Prof. Srinivasa Narasimhan

Carnegie Mellon University, U.S.

M.S. in Computer Vision, Robotics Institute

2022.8 – 2023.12

Advisor: Prof. Srinivasa Narasimhan

Wenzhou-Kean University, China

B.A. in Mathematical Sciences

2017.9 – 2021.6

Advisor: Prof. Gaurav Gupta

RESEARCH INTEREST

Computer Vision, Generative and World Models, Autonomous Driving, Embodied AI, Efficiency

SELECTED PUBLICATIONS

MapLightning: Online Vectorized HD Map Construction with 1D Map Tokens

Shen Zheng, Anurag Ghosh, Mani Ramanagopal, Srinivasa Narasimhan

Under Review [\[arXiv\]](#)

WarpI2I: Image Warping for Image-to-Image Translation

Shen Zheng, Anurag Ghosh, Gaurav Parmar, Srinivasa Narasimhan

ECCV 2026 [\[arXiv\]](#) [\[Page\]](#)

ROADWork: A Dataset and Benchmark for Learning to Recognize, Observe, Analyze and Drive Through Work Zones

Anurag Ghosh, **Shen Zheng**, Robert Tamburo, Juan R. Alvarez Padilla, Hailiang Zhu, Michael Cardei, Nicholas Dunn, Christoph Mertz, Srinivasa Narasimhan

ICCV 2025 [\[arXiv\]](#) [\[Page\]](#)

Instance-Warp: Saliency Guided Image Warping for Unsupervised Domain Adaptation

Shen Zheng*, Anurag Ghosh*, Srinivasa Narasimhan

WACV 2025 [\[arXiv\]](#) [\[Page\]](#)

TPSeNCE: Towards Artifact-Free Realistic Rain Generation for Deraining and Object Detection in Rain

Shen Zheng, Changjie Lu, Srinivasa Narasimhan

WACV 2024 [\[arXiv\]](#) [\[Page\]](#)

Low-Light Image Enhancement: A Comprehensive Survey and Beyond

Shen Zheng, Yiling Ma, Jinqian Pan, Changjie Lu, Gaurav Gupta

Survey [\[arXiv\]](#)

PointNorm: Dual Normalization is All You Need for Point Cloud Analysis

Shen Zheng, Jinqian Pan, Changjie Lu, Gaurav Gupta

IJCNN 2023 Oral [\[arXiv\]](#) [\[Page\]](#)

Semantic-Guided Zero-Shot Learning for Low-Light Image/Video Enhancement

Shen Zheng, Gaurav Gupta

WACV 2022 [\[arXiv\]](#) [\[Page\]](#)

AS-IntroVAE: Adversarial Similarity Distance Makes Robust IntroVAE

Changjie Lu, Shen Zheng, Zirui Wang, Omar Dib, Gaurav Gupta

ACML 2022 [\[arXiv\]](#)

INDUSTRY EXPERIENCE

Waymo

Perception Software Engineer Intern, 3D Occupancy and World Model

Summer 2026

Mentor: Zhenyao Zhu

Waymo

Perception Software Engineer Intern, Online HD Map Construction

Summer 2025

Mentor: Guohao Zhang

Lucid Motors

Perception Software Engineer, Traffic Light Detection

2024.2 – 2024.7

Director: Dr. Feng Guo

Momenta

Perception Engineer Intern, Traffic Light Detection

2021.9 – 2022.2

Director: Dr. Wangjiang Zhu